

MOHSEN MARZBAN

MSc ICT for Smart Societies · IoT & Embedded Systems ·
ML & Industry 4.0

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PROFESSIONAL SUMMARY

Final-year M.Sc. student in ICT for Smart Societies at Politecnico di Torino, combining 6+ years of hands-on industrial engineering experience (oil & gas, steel, cement) with academic expertise in Machine Learning, IoT system design, and smart infrastructure. Uniquely positioned to bridge Operational Technology (OT) and Information Technology (IT), with proven ability to design end-to-end data pipelines, deploy containerized IoT architectures, and apply ML models to real-world industrial and smart-society challenges.

TECHNICAL SKILLS

Programming	Python · MATLAB · SQL · Bash / Linux scripting
ML & Data Science	Scikit-learn · TensorFlow · NumPy · Pandas · Matplotlib · Regression · Classification · Neural Networks · Feature Engineering · Model Evaluation · ROC Analysis
IoT & Embedded	Raspberry Pi · Arduino · ESP32 · MQTT (Mosquitto) · BLE · LoRaWAN · REST APIs · Real-time sensor pipelines · Edge computing · Digital Twins · NetBird VPN
Cloud & DevOps	Docker · Docker Compose · AWS fundamentals · Azure basics · Git/GitHub · CI/CD basics · Linux process management
Databases	SQL Server · MySQL · MongoDB · InfluxDB
Tools & Platforms	Jupyter Notebook · Grafana · Power BI · EnergyPlus / OpenStudio · AutoCAD 2D/3D · Autodesk Navisworks · Microsoft Office Suite
Industrial / OT	P&ID · DCS/F&G · SCADA concepts · Instrumentation calibration & testing · Hi-pot/Megger testing · As-built drawings · PMBOK
Domain Knowledge	Smart Grids · Smart Buildings & BMS · Smart Mobility & ITS · IoT Security & Cryptography · GNSS & GIS · V2X Communications · Operations Research · Health Analytics

EDUCATION

M.Sc. — ICT for Smart Societies

Politecnico di Torino | Turin, Italy | 2023 – Present (Final Year)

Relevant coursework: Statistical Learning & Neural Networks · Mobile & Sensor Networks · ICT in Building Design (Digital Twins, BMS) · ICT for Smart Mobility (ITS, V2X, CRF Fiat collaboration) · Communication & Network Systems · Smart Grids (SCADA, SGAM, Blockchain) · Applied Cryptography & Cybersecurity · Operational Research · Machine Learning for Health · Satellite Positioning & GIS

B.Sc. — Electrical Engineering

Government-National University | Gorgan, Iran | 2015 – 2017

Associate Degree — Electrical Engineering

Government-National University | Yazd, Iran | 2013 – 2015

ACADEMIC PROJECTS

Smart Window Cyber-Physical System (WINECO)

Politecnico di Torino | ICT in Building Design — IoT & Edge Computing

- Engineered an automated smart window system using Raspberry Pi, Python, and multi-sensor arrays (BME680, CO₂, Lux) to regulate thermal comfort, indoor air quality, and lighting in real time.
- Architected a containerized IoT middleware stack using Docker Compose, orchestrating Eclipse Mosquitto MQTT broker, InfluxDB time-series storage, and KPI calculation microservices.
- Developed a priority decision engine integrating rule-based actuator control (stepper motors, PWM fans, LED modules) with external REST API calls (Open-Meteo) for predictive weather-based safety overrides.
- Deployed and tested the full system in a live university office environment, managing the complete lifecycle from SRR/PDR/CDR to hardware commissioning and real-environment validation.

Predictive Modeling of Parkinson's Disease Progression

Politecnico di Torino | Statistical Learning & ML for Health

- Built regression and Gaussian Process models to predict UPDRS neurological progression scores from clinical biomarker datasets using Scikit-learn, TensorFlow, Pandas, and NumPy.
- Performed end-to-end ML pipeline: data preprocessing, feature engineering, cross-validation, hyperparameter tuning, and ROC/sensitivity/specificity evaluation.

- Applied unsupervised clustering (K-Means, DBSCAN) and ICA-based EEG/ECG signal processing to support computer-aided diagnostic pipelines.

AI-Driven Building Energy Optimization & BMS Integration

Politecnico di Torino | ICT in Building Design

- Designed and optimized an energy-efficient building model using EnergyPlus and NSGA-II multi-objective optimization with surrogate ML models to minimize energy use and maximize indoor comfort (PMV/PPD, CO₂, lux).
- Integrated real-time BMS control components using MQTT, Docker, InfluxDB, and Grafana dashboards, applying white/grey/black-box modeling approaches.

Smart Grid Optimization & Blockchain Energy Trading

Politecnico di Torino | Smart Grids

- Modeled power flow dynamics in distribution topologies (IEEE 33 bus system) and implemented state estimation with SCADA/WAMS bad-data detection algorithms.
- Developed ML-based electricity price prediction models and evaluated blockchain (DLT) solutions for peer-to-peer energy trading in prosumer communities.
- Implemented cryptographic security protocols (AES, RSA, ECC, TLS/SSL, IPsec) and conducted threat modeling on IoT and cyber-physical infrastructure.

Network Optimization & Content Delivery Modeling

Politecnico di Torino | Operational Research & Network Management

- Formulated and solved LP/ILP optimization models for infrastructure resource allocation and capacity planning using Python optimization solvers.
- Designed metaheuristic algorithms (Simulated Annealing, Genetic Algorithms, Tabu Search) for NP-hard network routing and load-balancing problems.
- Modeled CDN traffic engineering using stochastic frameworks, queuing theory, and Markov chains to optimize QoS and cache hit-ratios.

PROFESSIONAL EXPERIENCE

MSc Thesis — Automated Computer Vision Pipeline for Damage Detection

3a Green Planet (3a S.r.l.) | Turin, Italy | Sep 2026 – Present

- Developing an end-to-end automation pipeline that replaces a manual inspection workflow using machine learning and computer vision.
- Implementing and comparing several deep-learning models for change detection and image segmentation (e.g. SegFormer, ChangeFormer, BIT) on real-world image data.
- Covering the full ML workflow: dataset preparation and annotation, model training, and performance evaluation.

Instrumentation & Electrical Technician

Maadkoush Steel / Behin Tarh Paydar / Farsnov Cement | Bandarabbas & Shiraz, Iran | Oct 2020 – Jul 2024

- Procured and quality-assured electrical and instrumentation components against industry standards, managing supplier coordination and project documentation.
- Conducted functional tests and inspections on field devices (transmitters, switches, valves), verifying compliance with technical specifications and sector regulations.

Electrical & Instrumentation Technical Office Engineer

Faradast Energy Falat — PGSOC (World's Largest Gas Condensate Refinery) / Control Poyan |

Bandarabbas, Iran | Jan 2018 – Jan 2020

- Analyzed complex engineering documents: P&IDs, DCS & F&G systems, loop diagrams, electrical schematics, and inspection reports in a high-stakes refinery environment.
- Installed and commissioned field instrumentation (transmitters TT/TE, pressure/level switches, indicators, valves) and performed calibration for measurement accuracy.
- Executed hi-pot and megger tests on cables; commissioned LV switchboards, motor starters (DOL), and lighting systems; created and maintained As-Built drawings.

LANGUAGES

Persian Native	English C1 — Proficient	Italian B1 — Intermediate
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I bring a rare combination: 6+ years of real-world engineering practice in high-stakes industrial environments (oil & gas, steel, cement) fused with an MSc curriculum built for the connected, intelligent world. I can read a P&ID and write a data pipeline. I understand both the sensor on the factory floor and the cloud infrastructure processing its data — making me immediately productive in IoT, ML, smart energy, mobility, and Industry 4.0 roles.

I consent to the processing of my personal data in accordance with GDPR 679/16.